THEORY OF EVOLUTION

Earth's history is marked by countless fascinating events that offer valuable insights into the origins and development of life. This module will explore the various theories of evolution, how evolution occurs, and the role of natural selection in shaping life as we know it.

Theories and Evidence of Evolution

We live in a world where we are not the only organisms that flourish. There are millions of other organisms with which we directly and indirectly interact. The number and kind of organisms living on Earth at a particular time is called **biological diversity.**

Evolution and Evolution Theory

Evolution is the change in species over time that shapes the diversity of life we see today. The **evolutionary theory** is the scientific explanation of this diversity. Evolution, or change over time, is the process by which present-day organisms have descended from ancient ones.

Characters of the individual organisms change. For example, in the case of Peacock, the lack of pigmentation caused by a mutation known as *leucism* produces the white peacock instead of iridescent blue, green, and brown tail feathers. Therefore, the organisms in a population do change.



Figure 1 & 2. Leucism – genetic mutation in Peacocks
Source: Sindhu, J. (2015). https://commons.wikimedia.org/wiki/File:Peacock_Plumage.jpg
Peacocks UK (2025) https://www.peacocksuk.com/peafowl-breeds/white-peacocks/

Paleontology and Geology

To provide facts about the various theories and evidence of evolution, two fields are involved: paleontology and geology.

Paleontology is the scientific study of past life, including the origin and extinction of organisms. **Paleontologists**, experts in this field, examine fossils to understand ancient ecosystems and how organisms evolved. For example, the fossil Archaeopteryx lithographica from the late Jurassic period provides evidence of the dinosaur ancestry of birds, displaying both dinosaur and bird-like features.



Figure 3. Archaeopteryx – the missing link between dinosaurs and birds
Source: Howells, C. and Buttler, C. (2011). https://museum.wales/articles/1011/Archaeopteryx---the-missing-link-between-dinosaurs-and-birds/

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Meanwhile, **Geology** is the study of Earth's history based on evidence found in rocks. **Geologists** contribute significantly to understanding the history of life on Earth by examining how the planet has changed over time, particularly due to shifts in climate and land formation.

Paleontologists and geologists are both working together to understand the history or evolution of life on Earth. Both of them use the geologic time scale to divide Earth's history into time intervals.

Development of Evolutionary Thought

The contributions of numerous scientists have shaped the understanding of how species change over time. Each thinker provided essential insights that laid the foundation for modern evolutionary theory. Below is a brief overview of key contributors and their contributions.

John Ray (1627-1705)
John Ray was an English naturalist who developed an early classification system for plants and animals. He introduced the concept of species, defining them as groups of organisms capable of interbreeding. His emphasis on observable characteristics influenced later taxonomic work.

Carl Linnaeus (1707-1778)
Known as the father of modern taxonomy, Carl Linnaeus developed the ***binomial nomenclature system***, which is still in use today. His classification system organized living organisms into hierarchical categories based on their physical traits. While he believed species were unchanging, his work provided a framework for studying biological relationships.

George Louis Leclerc, Comte de Buffon (1707-1788)
Buffon proposed that species could change over time, suggesting that the environment influences variations within organisms. His ideas challenged the notion of fixed species and paved the way for further exploration of evolutionary concepts.

Erasmus Darwin (1731-1802)
The grandfather of Charles Darwin, Erasmus Darwin, speculated that species evolved from common ancestors. In his writings, he suggested that competition and environmental factors drive species change, although he lacked a mechanism to explain how this occurred.

Georges Cuvier (1769-1832)
Cuvier was a French paleontologist who developed the ***theory of catastrophism***, which proposed that sudden, large-scale events shaped Earth's geological features. While he did not support the idea of evolution, his detailed study of fossils provided evidence of extinct species and contributed to the understanding of Earth's history.

Jean-Baptiste Lamarck (1744-1829)
Lamarck introduced one of the first coherent theories of evolution. He proposed that organisms adapt to their environments through acquired traits, which could then be passed on to offspring. Although his theory of inheritance of acquired characteristics was later disproven, Lamarck's emphasis on adaptation was influential.

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James Hutton (1726-1797)
James Hutton, known as the father of modern geology, introduced the concept of ***uniformitarianism***. He argued that Earth's geological processes occurred gradually over long periods, challenging the belief in a young Earth. His work laid the groundwork for understanding the immense timescales necessary for evolution.

Charles Lyell (1797-1875)
Building on Hutton's ideas, Lyell's principles of uniformitarianism further established that geological changes occur slowly and continuously. His writings influenced Charles Darwin by emphasizing the vast periods available for evolution to take place.

Charles Darwin (1809-1882)
Darwin formulated the **theory of natural selection** after his observations during the voyage of the HMS Beagle. He proposed that individuals with advantageous traits are more likely to survive and reproduce, leading to gradual changes in populations. His seminal work, *On the Origin of Species*, remains foundational to evolutionary biology.

Alfred Russel Wallace (1823-1913)
Wallace independently conceived the theory of natural selection. His extensive fieldwork in Southeast Asia and the Amazon provided evidence supporting the idea that species evolve through adaptation to their environments. Wallace's collaboration with Darwin helped solidify the acceptance of evolutionary theory. One of his significant contributions is the concept of the Wallace Line, a biogeographical boundary that separates the distinct animal species of Asia and Australia, highlighting the role of geographical barriers in species distribution and evolution.

These pioneering thinkers collectively advanced our understanding of how species change over time, shaping the foundation of evolutionary biology as we know it today.

Charles Darwin and His Theory of Evolution

Charles Darwin published the book "***On the Origin of Species by Means of Natural Selection***". Because of this, he is best known for his groundbreaking theory of evolution by *natural selection*. His ideas were shaped significantly by his observations during his voyage on the HMS Beagle, particularly in the Galapagos Islands.

Darwin's Observations in the Galàpagos Islands
In 1835, Darwin visited the Galapagos Islands, a volcanic archipelago located in the Pacific Ocean. During his time there, he studied the unique species of plants and animals, noting variations among the finches on different islands. These birds had different beak shapes and sizes, adapted to their specific diets.



Figure 4. Three Examples of Beak Variation in Galàpagos Finches
Source: Urry, L. A., Cain, M. L., Wasserman, S. A., Minorsky, P. V., & Reece, J. B. (2017). Campbell Biology (11th ed.) Pearson Education, Inc.

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Darwin hypothesized that these variations arose due to natural selection, where individuals with advantageous traits were more likely to survive and reproduce. This insight led to his formulation of the theory of evolution by natural selection. He also realized that the high birth rates and a shortage of life's basic needs would eventually force organisms into competition for resources. Thus, those who are stronger and faster can hunt for food and protect themselves from their enemies. This struggle for existence was vital to Darwin's theory of evolution.

Jean-Baptiste Lamarck and His Theory of Evolution

Jean-Baptiste Lamarck, a French naturalist, proposed one of the earliest theories of evolution. While his ideas have since been largely replaced by Darwin's theory, Lamarck's contributions were instrumental in advancing the concept of species change.

Lamarck's Theory of Use and Disuse

This theory suggests that organisms can change their traits during their lifetime based on the use or disuse of specific body parts. Frequently used structures become stronger and larger, while unused ones deteriorate over generations. For example, vestigial limb structures in snakes and pelvic bones in whales are remnants of their legged ancestors. Similarly, humans have vestigial structures like the appendix and tailbone. A well-known example is Lamarck's belief that giraffes once had short necks but stretched to reach tall leaves, causing their necks to elongate. He theorized that these acquired traits were passed to their offspring, resulting in longer necks over time.

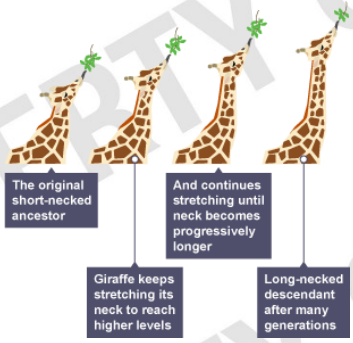


Figure 5. The elongation of the giraffe's neck, according to the theory of use and disuse
Source: BBC (n.d.) https://www.bbc.co.uk/bitesize/guides/zcqbdxs/revision/2

Theory of Inheritance of Acquired Characteristics

Lamarck's theory also posited that its offspring could inherit traits acquired during an organism's lifetime. For example, if an animal developed a muscular body from extensive physical activity, its offspring would inherit this trait.

Ancestral elephants, like the *Moeritherium*, were believed to have shorter trunks. Lamarck suggested that as these elephants stretched their trunks to reach food and water, their trunks gradually lengthened. This trait was then thought to be passed on to their offspring, resulting in elephants with progressively longer trunks over generations.

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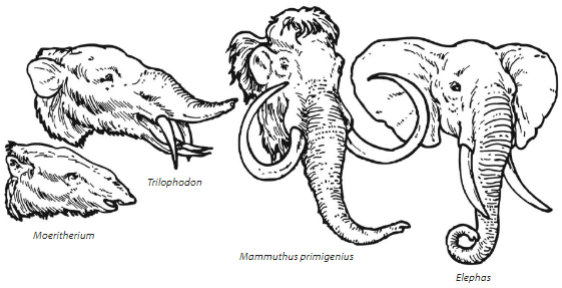


Figure 6. Evolution of Elephants.
Source: Mirabete, G. (2020). General Biology 2(Second Edition) e-Module. Diwa Learning Systems, Inc.

While this explanation is not supported by modern evolutionary science, it was an early attempt to describe how organisms adapt to their environments. Today, the development of the elephant's trunk is explained through natural selection, where longer trunks provide survival advantages, leading to their prevalence in later generations.

Evidence Supporting Evolution

Multiple lines of evidence demonstrate how organisms have changed over time, supporting the theory of evolution. These pieces of evidence are categorized as *direct* and *indirect*.

Direct Evidence of Evolution

This includes observable instances of evolution occurring within a relatively short time frame. These include:

- **Observable Evolutionary Changes**: Scientists have witnessed evolution in real-time, such as antibiotic resistance in bacteria and pesticide resistance in insects. These rapid adaptations show natural selection at work.

Indirect Evidence of Evolution

This includes clues that suggest common ancestry and evolutionary relationships among species. These include embryological, anatomical, and biochemical evidence.

- **Biogeography**
  The study of the distribution of species and ecosystems across the planet. It provides insights into how species adapt to their environments and how *geographic isolation* can lead to the formation of new species. This is where physical barriers such as mountains, oceans, or deserts prevent populations of a species from interbreeding. Over time, these isolated populations undergo genetic changes due to natural selection, resulting in the development of distinct species. This phenomenon is evident in island ecosystems, such as the Galápagos Islands, where Charles Darwin observed variations among finch species, each adapted to its specific environment.

  Furthermore, the concept of the Wallace Line — a boundary identified by Alfred Russel Wallace — highlights the distinction between species in the Asian and Australian regions, despite their geographical proximity. The stark difference in species on either side of the line supports the idea of evolution through environmental adaptation and geographical separation.

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Figure 7. The Wallace Line.
Source: National Geographic (2025). https://education.nationalgeographic.org/resource/dividing-species-wallace-line-map/

- **Fossil Evidence**
  Fossils provide historical records of extinct species and reveal evolutionary transitions. By dating fossils and studying their structures, scientists can trace how species have evolved. For example, the evolution of the horse is well-documented through fossils showing gradual changes in size, limb structure, and teeth adaptations over millions of years. Similarly, the discovery of *Tiktaalik*, a transitional fossil with features of both fish and early amphibians, provides evidence of the evolutionary shift from aquatic to terrestrial life.

- **Embryological Evidence**
  Embryology, the study of embryo development, reveals similarities among different species at early stages. Many vertebrates, including fish, reptiles, birds, and mammals, have similar embryonic features, such as gill slits and tail structures. These similarities suggest that these organisms share a common ancestor.

- **Anatomical Evidence**
  Comparing the physical structures of different organisms provides insights into evolutionary relationships. Three main types of anatomical evidence support evolution:
  - **Homologous Structures**: Body parts with similar structures but different functions (e.g., the human arm, batwing, and whale flipper) indicate common ancestry.

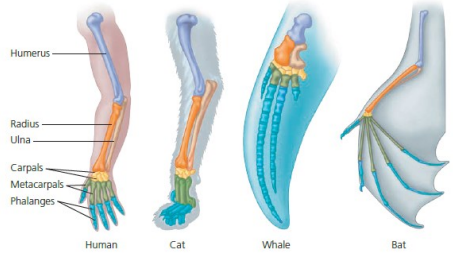


Figure 8. Mammalian forelimbs: homologous structures
Source: Source: Urry, L. A., Cain, M. L., Wasserman, S. A., Minorsky, P. V., & Reece, J. B. (2017). Campbell Biology (11th ed.) Pearson Education, Inc.

  - **Analogous Structures**: Structures with similar functions but different evolutionary origins (e.g., wings of birds and insects) show how other species can evolve similar adaptations through convergent evolution.

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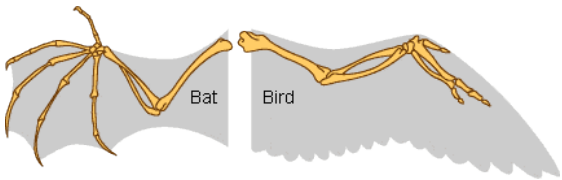


Figure 9. Analogous structure of the wings of bats and birds.
Source: Sanibel Sea School (2023) https://sanibelseaschool.org/blog/2020/06/24/homology-and-analogy-a-lesson-in-biology/.

- **Vestigial Structures**: Remnants of once-functional organs, such as the human appendix or the pelvic bones in whales, provide evidence that species have evolved and adapted over time.



Figure 10. Python with vestigial appendage (hind limb bud).
Source: American Museum of Natural History (n.d.) https://www.amnh.org/exhibitions/darwin/evolution-today/how-do-we-know-living-things-are-related/vestigial-organs

- **Biochemical Pieces of Evidence**
  Modern genetics and molecular biology provide strong evidence for evolution through similarities in DNA, RNA, and protein structures across species. Key examples include:
  - **Genetic Similarities**: Humans share a high percentage of DNA with other species, such as 98-99% similarity with chimpanzees, suggesting a recent common ancestor.
  - **Protein Comparisons**: The structure of proteins like cytochrome c, which plays a role in cellular respiration, is nearly identical across many species, indicating a shared evolutionary history.
  - **Universal Genetic Code**: All living organisms use the same genetic code, supporting the idea that all life descended from a common ancestor.

How Evolution and Natural Selection Occur

Natural Selection

In 1859, Darwin published On the Origin of Species, introducing two main ideas: descent with ***modification*** and ***natural selection***. He proposed that all species share a common ancestor and gradually change over time through inherited traits. As genetic variations arise during reproduction, beneficial traits accumulate over generations, leading to new species and the diversity of life we see today. Over time, this results in the accumulation of favorable characteristics. An organism's **fitness**, or its ability to survive and reproduce in a particular environment, is a central factor in natural selection. Individuals with poor traits suited to their surroundings are less likely to survive and produce offspring, while well-adapted individuals thrive.

Darwin's concept of natural selection explains how certain traits become more common in a population. He had made two (2) notable observations, from which he drew two interferences:

- **Observation #1**: Members of the Population often vary in their inherited traits.
- **Observation #2**: All species can produce more offspring than their environment can support, and many of these offspring fail to survive and reproduce.
- **Inference #1**: Individuals whose inherited traits give them a higher probability of surviving and reproducing in a given environment tend to leave more offspring than other individuals.

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- **Inference #2:** This unequal ability of individuals to survive and reproduce will lead to the accumulation of favorable traits in the population over generations.

Mechanisms of Natural Selection

Natural selection operates through several key mechanisms that determine which traits are passed on to future generations based on their impact on survival and reproduction. It is divided into several parts that support how evolution happens.

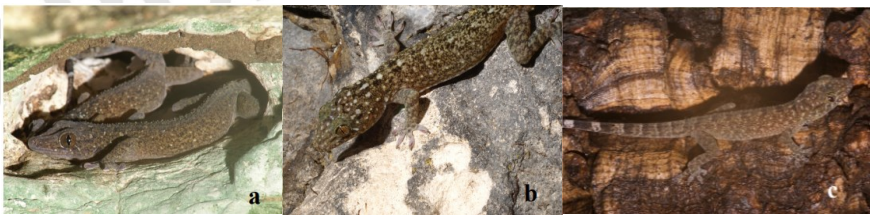
1. Overpopulation.
Darwin, influenced by Thomas Malthus, noted that organisms produce far more offspring than their environment can support. This overproduction leads to competition for limited resources. For example, the Green Sea Turtle (*Chelonia mydas*) lays around 100 eggs per nest, but few hatchlings survive due to predation, harsh conditions, and human activities. Without these limiting factors, turtle populations would grow uncontrollably, overwhelming the ecosystem.



Figure 11. A mother turtle laid 120 eggs at dawn in Dahican, Davao Oriental.
Source: DENR Davao (2020)

2. Struggle for Existence
Limited resources create competition among organisms for survival. This struggle can be direct, like predators fighting for prey, or indirect, like plants competing for sunlight and water. In Mindoro, the critically endangered tamaraw (*Bubalus mindorensis*) competes with livestock for grazing areas and water sources. Similarly, the Philippine Crocodile (*Crocodylus mindorensis*) faces threats from habitat destruction and harmful fishing practices, leaving fewer than 137 mature individuals in the wild.

3. Presence of Variation
Genetic variations among individuals occur through mutation or recombination, enhancing survival chances. These differences allow organisms to adapt to their environment. For instance, the Panay Limestone Gecko (*Gekko ernstkelleri*) from Antique exhibits distinct sepia-colored spots, helping it blend into its surroundings for protection against predators. Such variations increase the likelihood of survival and reproduction.



Figures 12, 13 & 14. Gekko ernstkelleri, Panay Island
Source: Gaulke, M., and Demegillo, A. (2020). Amphibian and reptile species described from Panay Island, Philippines. Philippine Journal of Systematic Biology.

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4. Survival of the Fittest

Organisms with advantageous traits have a higher chance of survival and reproduction. While cheetahs are the fastest land animals, gazelles can sustain their speed over longer distances, increasing their chances of escape. Similarly, a vibrant lotus flower is more likely to attract pollinators, ensuring reproductive success. Further, this explains how beneficial traits become more common in a population.

5. Transmission of Variations

Favorable traits that enhance survival and reproduction are passed down to offspring, leading to adaptation over generations. Traditional rice varieties in the Philippines have developed resistance to pests and flooding through selective breeding. By planting seeds from resilient crops, farmers ensure the inheritance of advantageous traits, improving agricultural sustainability.

Types of Natural Selection

There are three main types of natural selection: stabilizing selection, directional selection, and disruptive selection.

1. Stabilizing Selection

A type of natural selection that favors individuals with average traits while selecting against extreme traits, reducing variation within a population. This typically occurs in stable environments where extremes are disadvantageous. For example, human birth weight demonstrates stabilizing selection, as babies with average weights have higher survival rates, while those with very low or high weights face greater health risks.

2. Directional Selection

This occurs when individuals with traits at one extreme are favored, causing a shift in the population's traits over time. This often results from environmental changes. A well-known example is the peppered moth during the Industrial Revolution in England. Darker moths, better camouflaged against soot-darkened trees, had a survival advantage, leading to an increase in their population.

3. Disruptive Selection

This is also known as ***diversifying selection,*** which favors individuals with extreme traits at both ends of the spectrum, while those with intermediate characteristics are disadvantaged. It can increase genetic diversity and potentially lead to speciation. In some bird populations, those with very large or small beaks may efficiently access different seed types. In contrast, birds with medium-sized beaks struggle, resulting in a bimodal distribution of beak sizes.

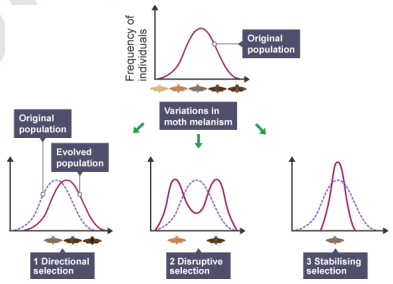


Figure 15. Types of Natural Selection

Source: BBC (n.d.) https://www.bbc.co.uk/bitesize/guides/zmf9xyc/revision/4

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Artificial Selection, Non-Random Mating, Inbreeding, Outbreeding, and Genetic Drift

Evolutionary changes in populations are not solely driven by natural selection. Several other mechanisms play a role in shaping genetic diversity and the traits observed within populations.

Artificial Selection

This refers to the intentional breeding of organisms with desirable traits by humans. Unlike natural selection, which occurs through environmental pressures, artificial selection is guided by human preferences to enhance specific characteristics. For example, farmers breed livestock for higher milk production or select crops for improved yield and disease resistance.

Non-Random Mating

This occurs when individuals select mates based on specific traits rather than mating randomly. This can lead to changes in allele frequencies and influence the genetic makeup of a population. Peacocks with more vibrant plumage are often preferred by females, leading to the prevalence of bright feathers in future generations.

Inbreeding is the mating of closely related individuals within a population. While it can maintain desired traits, it also increases the likelihood of genetic disorders by amplifying recessive alleles.
Further, **Outbreeding** involves mating between unrelated or distantly related individuals, which enhances genetic diversity and reduces the risk of inherited diseases. It often results in offspring with greater resilience and fitness. For instance, crossbreeding different crop varieties to produce hybrid plants with superior traits.

Genetic Drift

It refers to random changes in allele frequencies within a population, particularly in small populations. Unlike natural selection, genetic drift is not driven by adaptive advantage but by chance events. As an example, a natural disaster may randomly eliminate a portion of a population, leading to changes in genetic diversity.

A **population bottleneck** occurs when a large proportion of a population is suddenly reduced due to an environmental event such as a natural disaster, disease outbreak, or human activity. This drastic reduction in population size results in a loss of genetic diversity.

Meanwhile, the **founder effect** occurs when a small group of individuals establishes a new population, leading to reduced genetic variation compared to the original population. The genetic traits of the founding members disproportionately influence the gene pool of the new population.

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